

CPEN 416 / EECE 5710

Lecture 13: QFT and quantum phase estimation

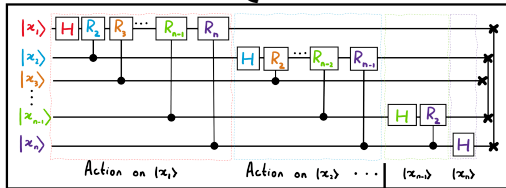
Thursday 23 October 2025

Announcements

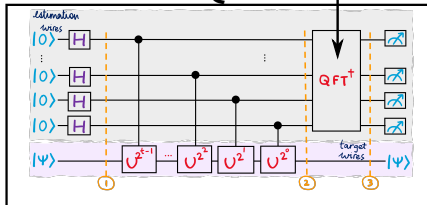
- Quiz 6 on Tuesday
- Tutorial Assignment 2 due tomorrow 23:59
- Tutorial Assignment 3 next week
- Practice Assignment 3 distributed tomorrow

Reminder: where are we going?

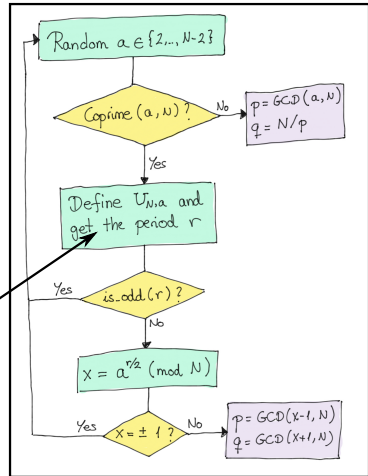
1. QFT



2. QPE



3. Shor



Last time

We introduced the quantum Fourier transform, which is analogous to the inverse discrete Fourier transform.

$$QFT|x\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{xk} |k\rangle$$

$$QFT = \frac{1}{\sqrt{N}} \begin{pmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \omega & \omega^2 & \dots & \omega^{N-1} \\ 1 & \omega^2 & \omega^4 & \dots & \omega^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{N-1} & \omega^{2(N-1)} & \dots & \omega^{(N-1)(N-1)} \end{pmatrix}$$

where for n qubits, $N = 2^n$, and $\omega = e^{2\pi i/N}$

Last time

We worked out the result of the QFT for a few small cases:

$$QFT|x_1\rangle = \frac{1}{2} (|0\rangle + e^{2\pi i 0.x_1}|1\rangle)$$

$$QFT|x_1x_2\rangle = \frac{1}{2} \left((|0\rangle + e^{2\pi i 0.x_2}|1\rangle) \otimes (|0\rangle + e^{2\pi i 0.x_1x_2}|1\rangle) \right)$$

Example: verify with $QFT|01\rangle$

Last time

$$QFT|x_1x_2x_3\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 0.x_3}|1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 0.x_2x_3}|1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 0.x_1x_2x_3}|1\rangle)$$

General expression:

$$\begin{aligned} |x\rangle &\rightarrow \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{xk} |k\rangle \\ &= \frac{(|0\rangle + e^{2\pi i 0.x_n}|1\rangle) (|0\rangle + e^{2\pi i 0.x_{n-1}x_n}|1\rangle) \dots (|0\rangle + e^{2\pi i 0.x_1 \dots x_n}|1\rangle)}{\sqrt{N}} \end{aligned}$$

Learning outcomes

- Construct a circuit for the quantum Fourier transform and analyze its resource usage
- Outline the algorithmic steps of the QPE subroutine
- Implement QPE in PennyLane and estimate the phase of small unitary matrix

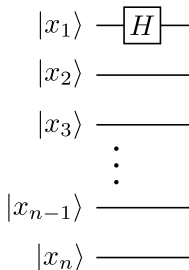
A circuit for the QFT

$$|x\rangle \rightarrow \frac{(|0\rangle + e^{2\pi i 0 \cdot x_n} |1\rangle) (|0\rangle + e^{2\pi i 0 \cdot x_{n-1} x_n} |1\rangle) \cdots (|0\rangle + e^{2\pi i 0 \cdot x_1 \cdots x_n} |1\rangle)}{\sqrt{N}}$$

Starting with the state

$$|x\rangle = |x_1 \cdots x_n\rangle,$$

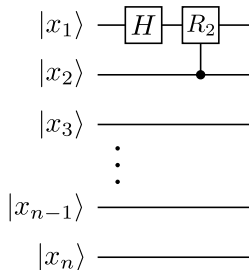
apply a Hadamard to qubit 1:



A circuit for the QFT

$$|x\rangle \rightarrow \frac{(|0\rangle + e^{2\pi i 0.x_n}|1\rangle)(|0\rangle + e^{2\pi i 0.x_{n-1}x_n}|1\rangle) \dots (|0\rangle + e^{2\pi i 0.x_1 \dots x_n}|1\rangle)}{\sqrt{N}}$$

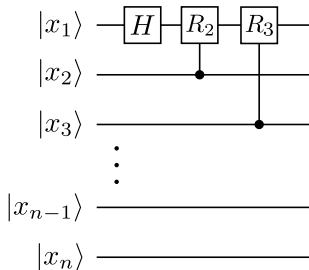
Apply controlled R_2 from qubit
2 $\rightarrow$ 1



A circuit for the QFT

$$|x\rangle \rightarrow \frac{(|0\rangle + e^{2\pi i 0 \cdot x_n} |1\rangle) (|0\rangle + e^{2\pi i 0 \cdot x_{n-1} x_n} |1\rangle) \dots (|0\rangle + e^{2\pi i 0 \cdot x_1 \dots x_n} |1\rangle)}{\sqrt{N}}$$

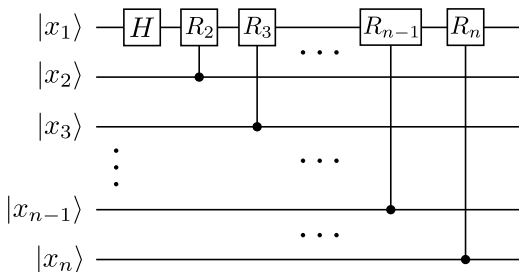
Apply controlled R_3 from qubit
3 $\rightarrow$ 1



A circuit for the QFT

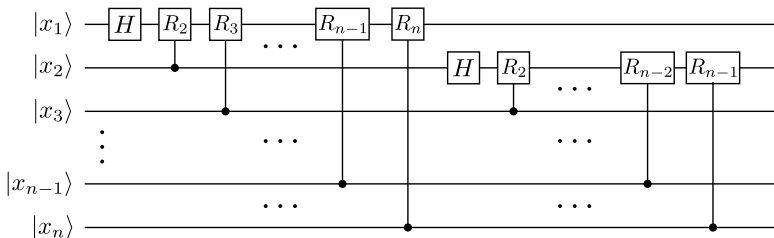
$$|x\rangle \rightarrow \frac{(|0\rangle + e^{2\pi i 0 \cdot x_n} |1\rangle) (|0\rangle + e^{2\pi i 0 \cdot x_{n-1} x_n} |1\rangle) \dots (|0\rangle + e^{2\pi i 0 \cdot x_1 \dots x_n} |1\rangle)}{\sqrt{N}}$$

Apply a controlled R_4 from $4 \rightarrow 1$, etc. up to the n -th qubit to get



A circuit for the QFT

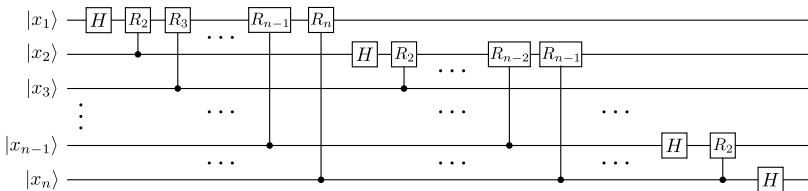
Similar thing with second qubit: apply H then controlled rotations from qubits 3 to n to get



A circuit for the QFT

Do this for all qubits to get that big ugly state from earlier:

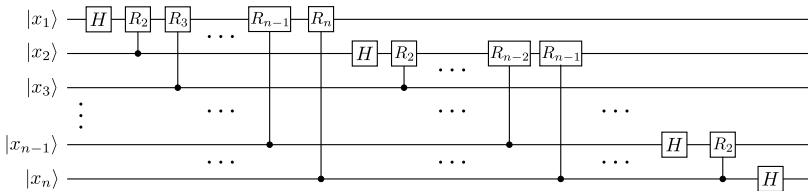
$$|x\rangle \rightarrow \frac{(|0\rangle + e^{2\pi i 0 \cdot x_n} |1\rangle) (|0\rangle + e^{2\pi i 0 \cdot x_{n-1} x_n} |1\rangle) \dots (|0\rangle + e^{2\pi i 0 \cdot x_1 \dots x_n} |1\rangle)}{\sqrt{N}}$$



Qubit order is backwards - easily fixed with SWAP gates.

Quantum Fourier transform

Exercise: What are the gate counts and depth of this circuit?



Eigenvalues of unitary matrices

Eigenvalues of unitary matrices are complex numbers with magnitude 1, i.e.,

where θ_k is a phase, $|\theta_k| \leq 1$. (See proof at end of slides)

What if we want to *estimate* an unknown θ_k ?

Quantum phase estimation

Given unitary U and eigenvector $|k\rangle$, estimate θ_k such that

$$U|k\rangle = e^{2\pi i\theta_k}|k\rangle$$

Must determine:

- How to design a circuit that extracts θ_k
- What precision can we estimate it to
- What to do if we don't know $|k\rangle$ in advance

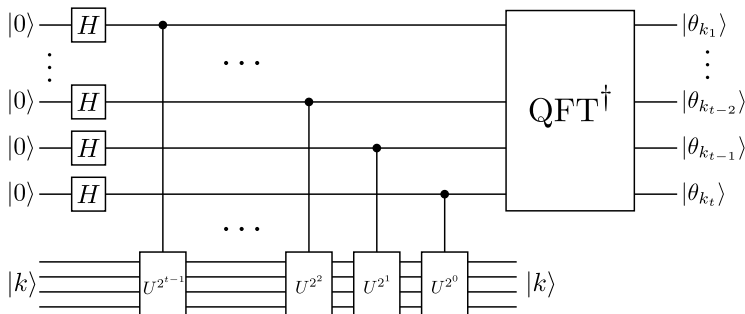
(You can explore the last two in practice assignment 3!)

Quantum phase estimation

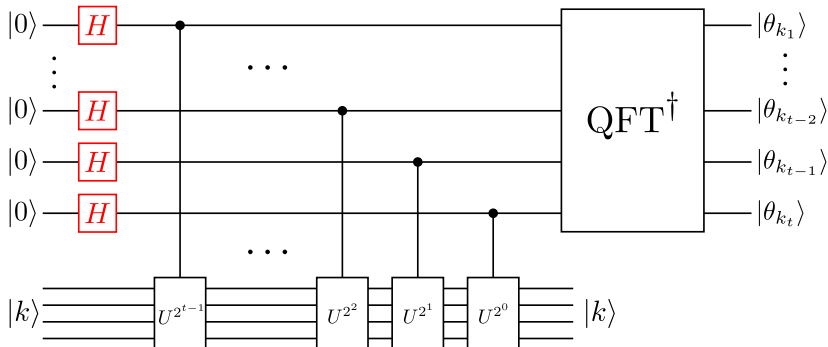
Let U be an n -qubit unitary; $|k\rangle$ is an n -qubit eigenstate.

Assume θ_k can be represented *exactly* using t bits:

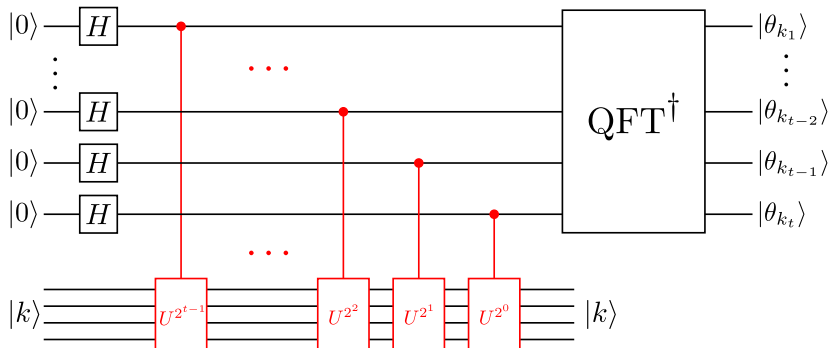
$$\theta_k = 0.\theta_{k_1} \cdots \theta_{k_t}$$



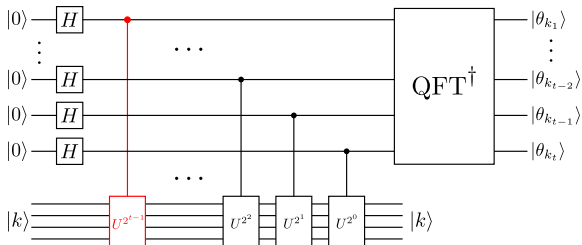
Quantum phase estimation: step 1



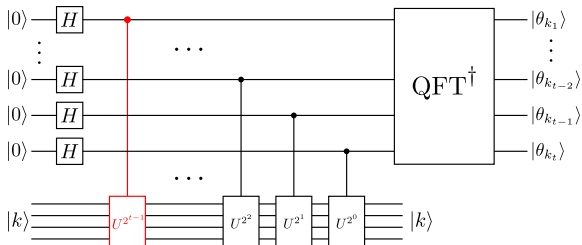
Quantum phase estimation: step 2



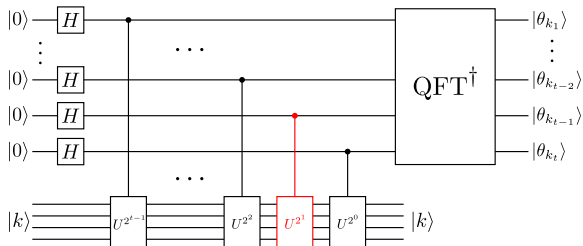
Quantum phase estimation: step 2



Quantum phase estimation: step 2

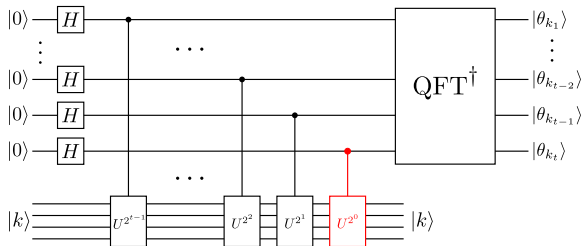


Quantum phase estimation: step 2



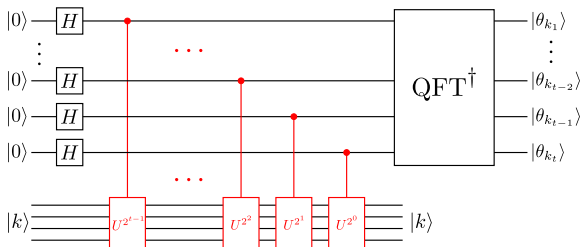
Check second-last qubit (ignore the others)

Quantum phase estimation: step 2



Can show in the same way for the last qubit (ignore others)

Quantum phase estimation: step 2



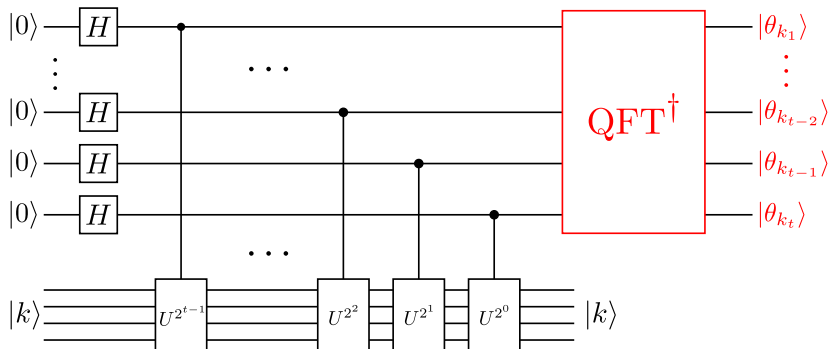
After step 2, we have the state

$$\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 0 \cdot \theta_{k_t}} |1\rangle) \cdots \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 0 \cdot \theta_{k_2} \cdots \theta_{k_t}} |1\rangle) \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 0 \cdot \theta_{k_1} \cdots \theta_{k_t}} |1\rangle) |k\rangle$$

Should look familiar!

Quantum phase estimation: step 3

Measure to learn the bits of θ_k .



Example: QPE for the T gate

Let's apply QPE to estimate the phase of an eigenstate of T : $|1\rangle$.

- ❶ What answer do we expect?
- ❷ How many estimation bits?
- ❸ What does the circuit look like?

Quantum phase estimation

Recall our initial questions... any insights about the second two?

- How to design a circuit that extracts θ_k
- What precision can we estimate it to
- What to do if we don't know $|k\rangle$ in advance

Next time

Content:

- Tutorial assignment 3 (QAE with QPE)
- Quiz 6 on Tuesday
- Order finding

Action items:

- ① Work on project (do peer assessment)
- ② Finish tutorial assignment 2

Recommended reading:

- From today: Codebook QFT, QPE, Nielsen & Chuang 5.1-2
- For next class: Codebook SH, Nielsen & Chuang 5.3.1

Eigenvalues of unitary matrices

Fun fact: eigenvalues of unitaries are complex numbers with magnitude 1.

Proof: Let λ_k be the eigenvalue associated with eigenvector $|k\rangle$ of a unitary U :

$$U|k\rangle = \lambda_k|k\rangle$$

Take the conjugate transpose:

$$\langle k| U^\dagger = \langle k| \lambda_k^*$$

Multiply the two sides together:

$$\langle k| U^\dagger U|k\rangle = \langle k| \lambda_k^* \lambda_k |k\rangle$$

$$\langle k|k\rangle = |\lambda_k|^2 \langle k|k\rangle$$

$$1 = |\lambda_k|^2$$